

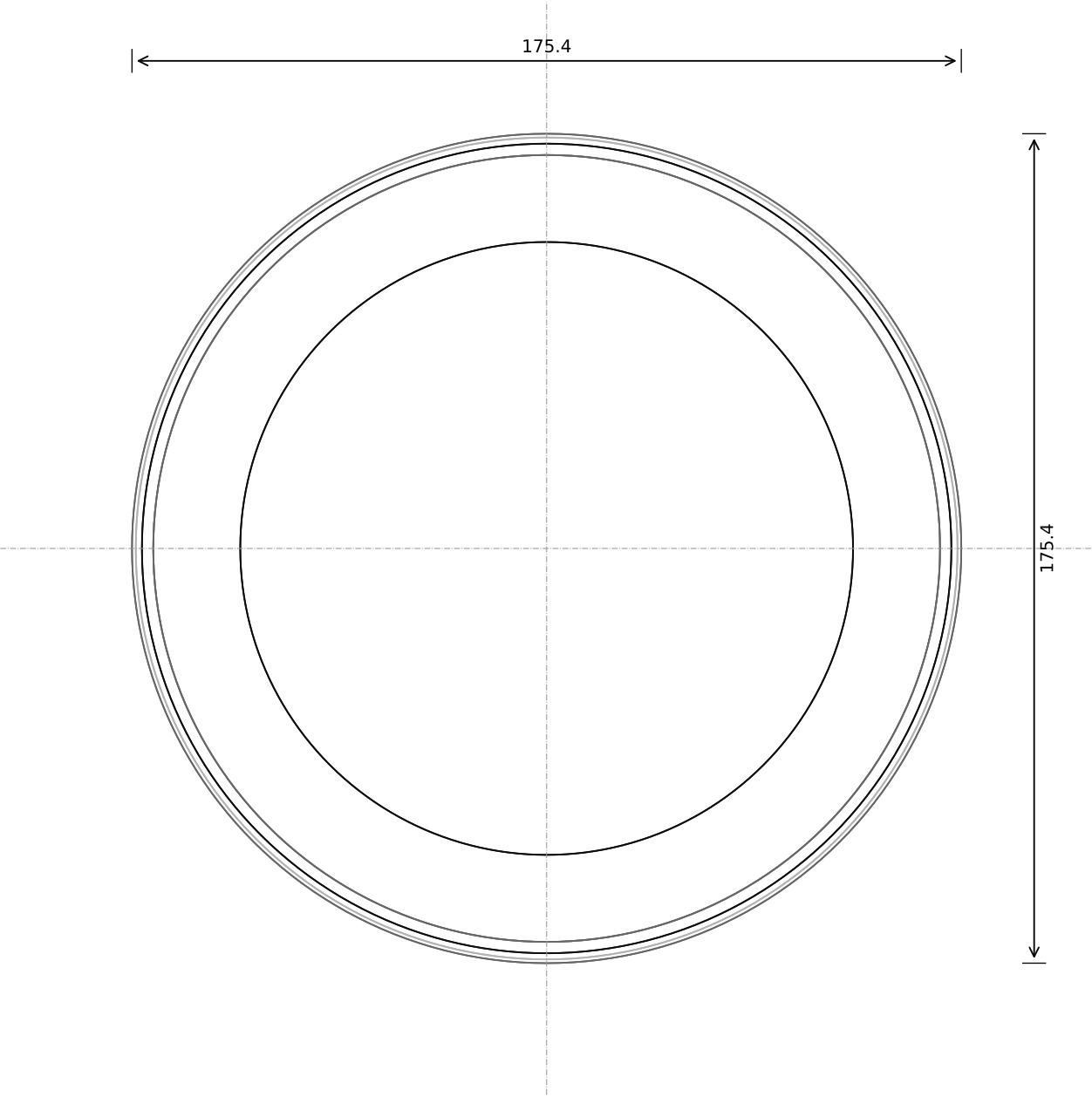
the 60 — v15 — sheet 0: the ASSUMED numbers (every one of these must be verified on the real part before the drawing is trusted)

2026-09-06. Stack: plate 8.0, adapter 14.78-16.38 (socket to 23.68), code ring under the crown at 31.53, seat 26.0, glass 26.50-28.48, lens to 30.98, knob top 34.38, overall 35.88 with the pad. Ø175.

1. cover lens: Ø140.0 x 2.5	specified separately; carrier rim and lip overlap designed to it
2. bonding tape: 0.5	double-sided foam tape between the seat and the glass edge (RULING 5 Sep: glue allowed for the display)
3. panel back component area: 60.0 x 68.36 x 1.5 at -14.0	position from the drawing's back view
4. Active Cooler position: from Pi x 2.5, fan from x 43.0	no CAD body; footprint and fan position from photographs
5. cooler base above the board: 2.8	SoC 2.4 + thermal pad
6. fan intake plenum: 8.0	minimum free height above the blower
7. micro-HDMI receptacles: x 26.0/39.5	not in the vendor STEP; drawn as envelopes
8. HDMI ribbon plugs: 12x8x5 micro end, 21x12x6 A end	flat FPC HDMI cable, 200 mm
9. adapter board thickness and part heights: 1.6; parts 2.5, underside 1.0, flex connector 2.0	outline, holes, socket size and position PUBLISHED (datasheet drawing)
10. display connect board: 30.0 x 46.0 x 1.2, connectors 2.0	part of the adapter kit; size scaled from its photograph, hole positions assumed at the corners
11. Pi standoffs and washers: M2.5 x 12.0 hex, 0.5 washers	the adapter stack: closing-plate land / washer / Pi / standoff / adapter
12. USB-C up-angle plug: 12 x 12.5 x 6.5 head	measure the lead
13. audio / motion / touch / converter boards: envelopes	electrical layout not done
14. barrel jack body: 14.4 x 9.0 x 11.0 on a vertical board	PJ-063AH; CAD body to fetch
15. right-angle barrel plug: Ø8.5 body	measure the brick's plug
16. MOTOR_BASE_OD / MOTOR_BASE_H: 30.0 / 4.3	carried from v9
17. MOTOR_BAND_T: 0.6	carried from v9
18. servo pushrod: 3 mm bridge to the tab	carried from v9; servo on the plate in a frame
19. LED strip: 5.0 wide x 1.6 thick	any 5 mm addressable strip; a wider strip raises the halo band
20. blower inlet and outlet: inlet Ø24.0 in the face, outlet 21.0 x 7.0 from 1.5 up	from the datasheet drawing (outline, holes and leads PUBLISHED)
21. blower gaskets: 0.3 closed-cell foam	under the inlet face and the hood
22. closing-plate gasket: 0.2 die-cut	in a groove of the recess floor, on every sealing wall
23. Pi 5 fan connector current rating: 0.13 A drawn (0.20 max)	the connector sits in the vendor STEP; its rating is not published
24. encoder shim family: [0.7, 0.8, 0.9, 1.0, 1.1]	printed; the height is chosen on the bench

Build-time changes against V11-SPECIFICATION.md are listed in V11.md. Sheets 1-11 follow: one per made part.

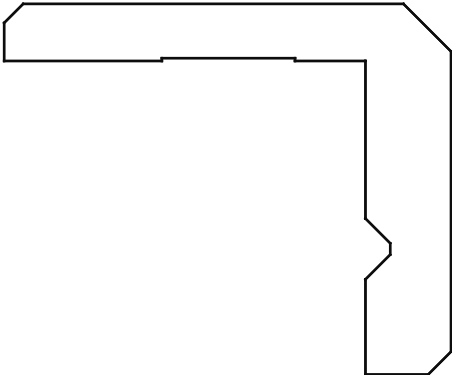
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 58 to 90, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. outside Ø175.4, bore Ø166.4 (0.45 to the panel's chin corners); wall 4.5
- 2. top rim: lip inner edge Ø128.4, 1.0 inner chamfer, 20 flat, 2.5 outer chamfer (RULING 6 Sep: rim >= 20); lip 3.0 thick, top at z 34.38
- 3. outer chamfer 2.5 x 45° smooth; inner chamfer 1.0; skirt chamfer 1.2
- 4. V-groove 90°, root r 84.50399999999999, z 19.9-23.1
- 5. code RING recess 0.15 deep in the crown's underside, r 72.5-79.5 (v13: the encoder reads the crown, not the bore - the bore is plain apart from the V-groove)
- 6. diamond knurl 78 starts x 2, 1.0 deep, helix 36.8°, 3 whole rows z 16.9-31.1; skirt bottom z 14.9
- 7. wall under the knurl root 3.5

the 60 — v15 Knob

Cadrane · sheet 1 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / CNC 6082-T6, hard anodised (production)

Process: print top face down, 0.16 mm; no supports

mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

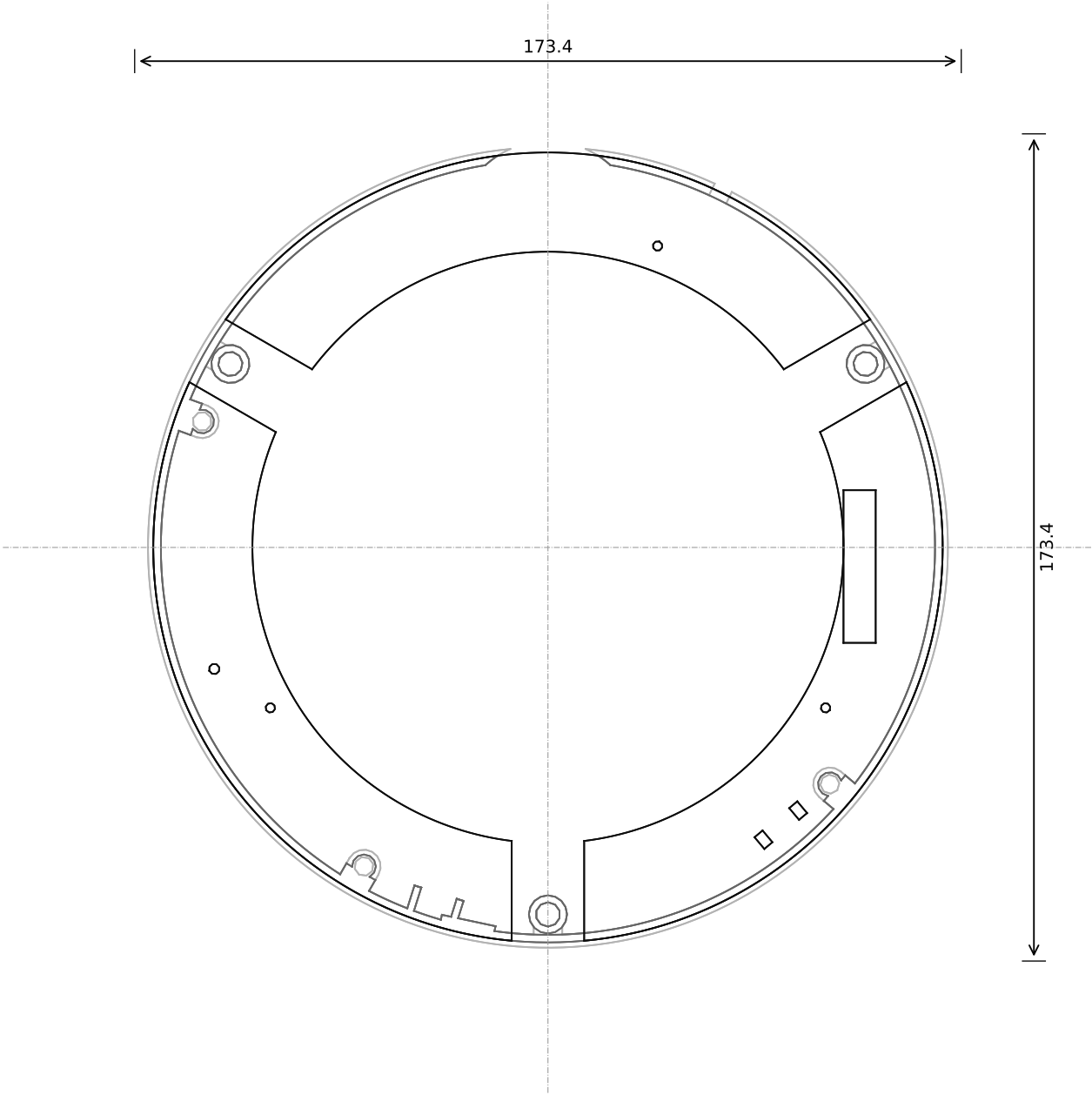
Part: knob_body

Overall 175.4 x 175.4 x 19.5

Volume 69.36 cm³

Scale: see each view (fit)

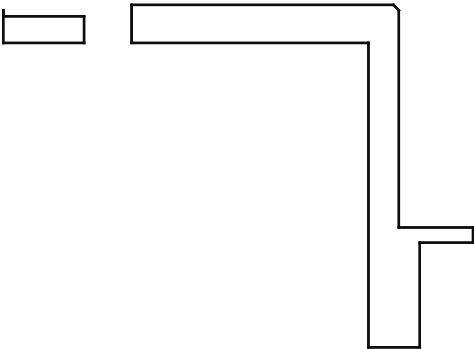
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 57 to 89, enlarged)



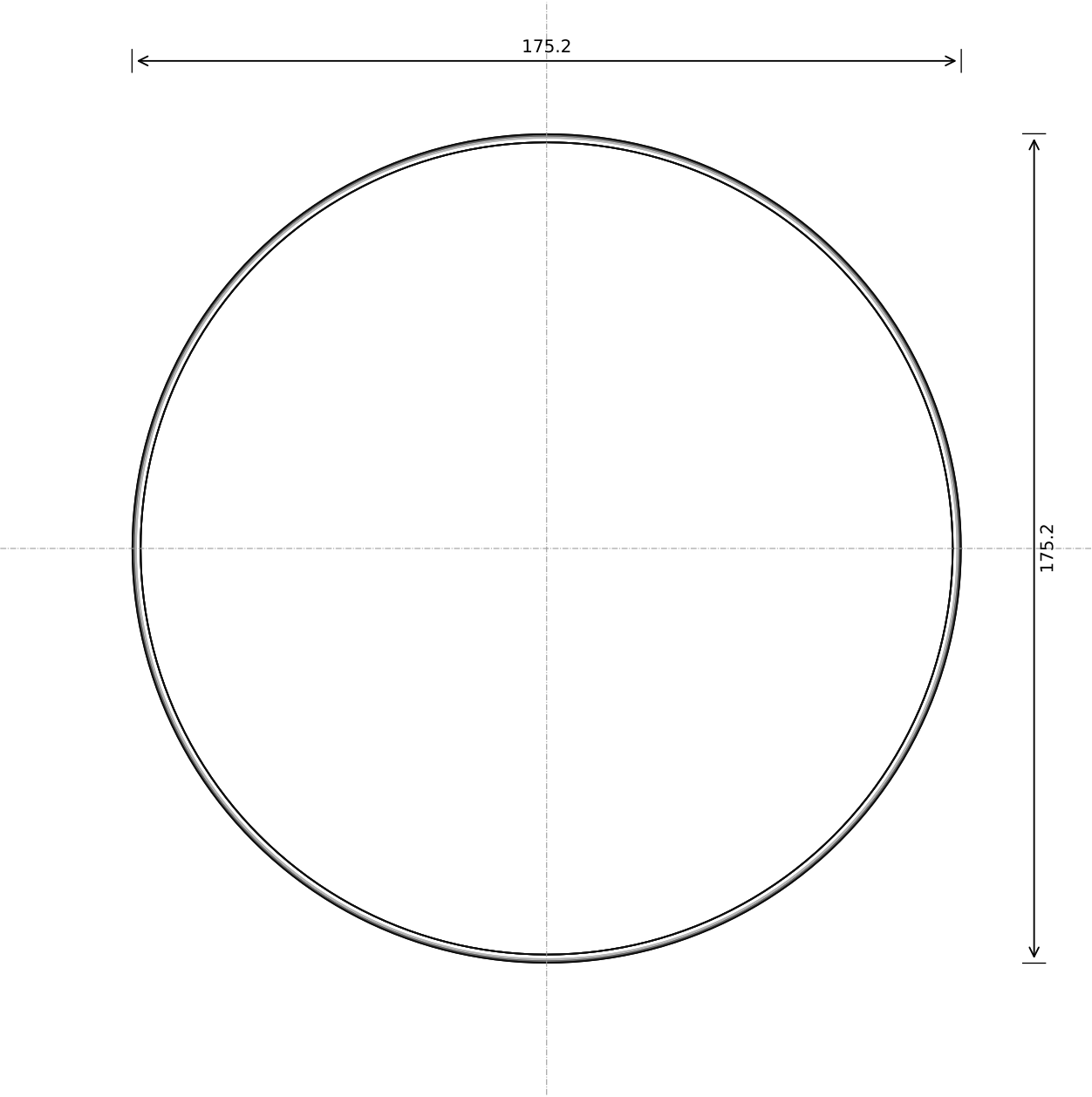
SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. wall r 81.204-82.80399999999999, z 8.0-26.0	the 60 — v15	Internal structure
2. seat flange r 62.0-82.80399999999999, z 24.0-26.0: its top face is FLAT (nothing rises above it) with 0.3 chamfers on both edges; three Ø2.0 H7 through-holes (ream) at [70, 210, 330]°, r 67.25 for the display's dowel pins, each in a Ø5.0 boss 4.0 under the flange	Cadrane · sheet 2 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)	
3. LED-strip band r 82.80399999999999-83.90399999999998, z 8.0-13.5 (strip r 84.00399999999998-85.60399999999997, 53 LEDs, 2.54 A full white); diffuser lip r 82.80399999999999-86.704, z 13.5-14.3	Material: PETG (prototype) / PA12 (production)	Part: internal_structure
4. wheel posts Ø8.0 at r 76.93, az [30, 150, 270], to z 18.9; bush bores Ø5.1; wall windows 15.2 wide, open to the top from z 18.6; the seat flange is cut away above each wheel	Process: v15: print INVERTED - the seat flange's top face on the bed, 0.20 mm support, 4x1.734x18.01	Overlap: 173.4 on 173.4, extent 18.01
5. pillars Ø7 at r 77.204, az [160, 240, 320], to z 16.0; M3 inserts from below (the plate screws) - nothing screws in from above	the flat top, sensor up, 2.0 under the crown's code ring; LPA pad at 258.0° with the wpp board, dip beside its flex tail at 252.5° (two ribs r 76-81.7, a 0.4 mm, tolerance ±0.1 unless stated, clearances: running 0.30,	Volume 35.70 cm³
6. encoder at 310.0°, r 76: two radial slots 2.2 x 3.2 through the flange at t ±4.75 over M2 inserts in Ø5.0 bosses under it (the separate 0.9 shim and the 8 x 12 board sit on the	static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)
7. flex slot 32 x 2.5 through the seat at 0°, offset -4; 101 vent slits 1.0 x 4.5 at z 18.5-23.0, 2.5° grid		
8. motor relief at 90°		

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 58 to 90, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. r 85.8-86.7 at the bottom, to r 87.7 at the top; z 8.0-13.5
- 2. 0.5 chamfer on the outside top edge
- 3. retained by the structure's lip (r to 86.7); sits on the rim ring 0.2 outside the LED strip

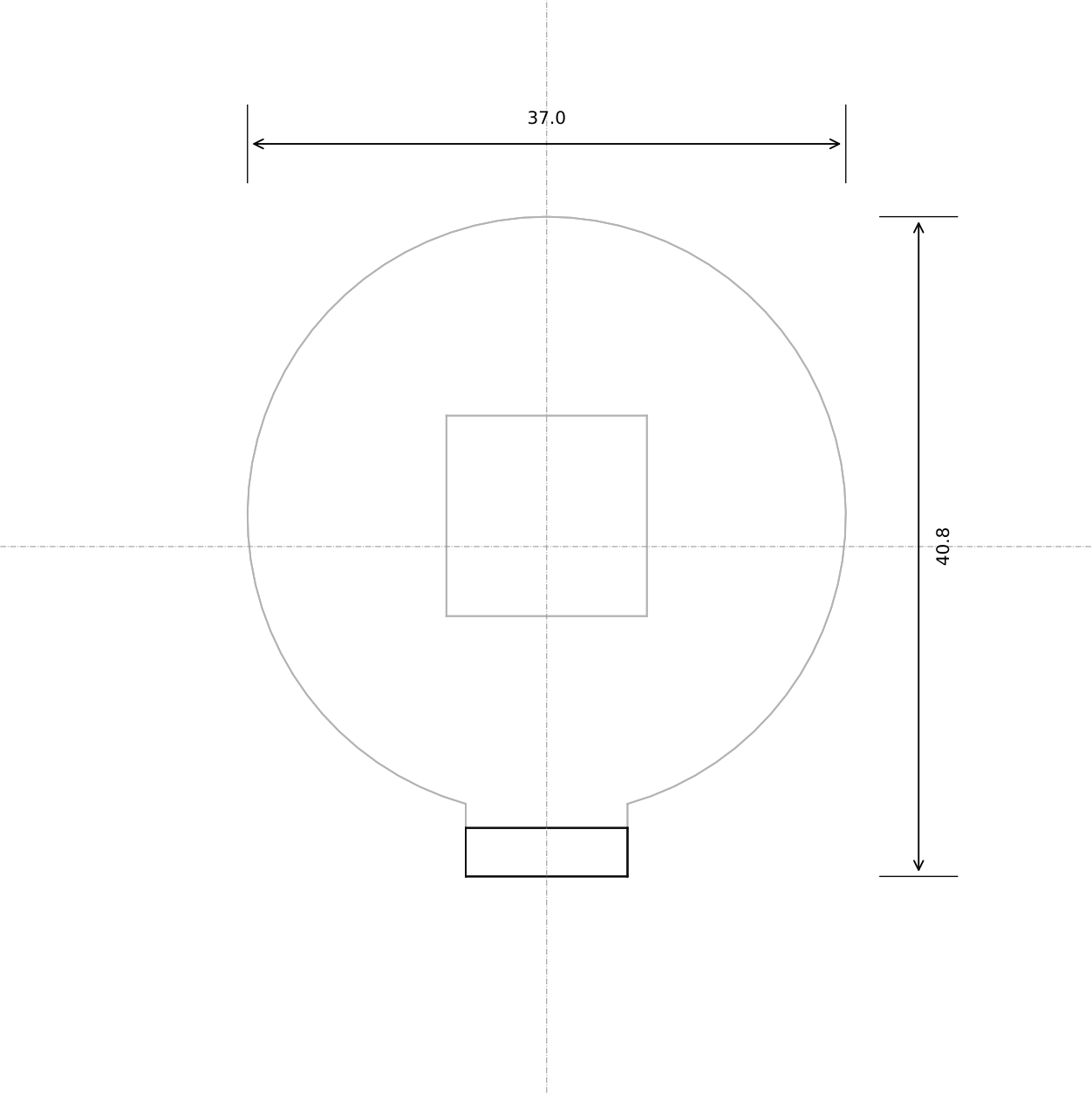
the 60 — v15Halo diffuser

Cadrane · sheet 3 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

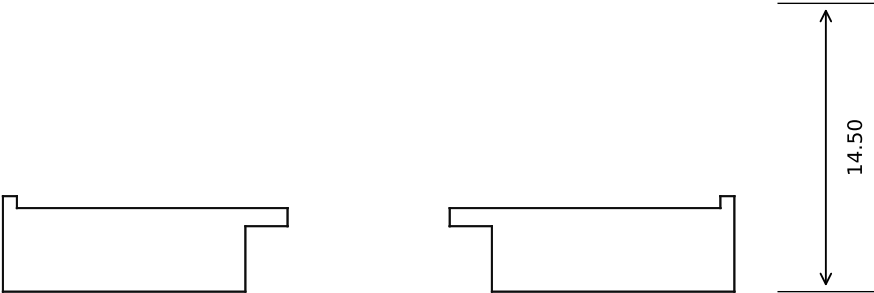
Material: opal PMMA (production) / natural PETG (prototype)
Process: print on its underside, 0.20 mm; no supports
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: halo_diffuser
Overall 175.2 x 175.2 x 5.5
Volume 4.12 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

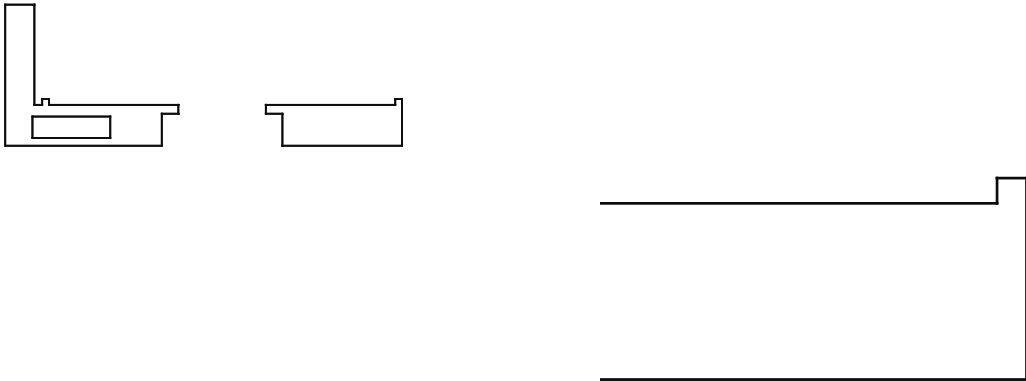


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 9 to 20, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. shoe $\varnothing 37 \times 4.2$; locating rim $\varnothing 35.6/37 \times 0.6$
- 2. commutation-board pocket $12.4 \times 12.4 \times 3.3$; sight hole $\varnothing 9$; v14: a 2.4×2.2 notch through the shoe's inboard wall for the sensor lead
- 3. push tab 3.0×10.0 to $z\ 14.5$ (v11: 28.5 through the deck; the servo is now beside it on the plate)
- 4. travel 2.4 radial in the plate's stadium hole

the 60 — v15

Motor carriage

Cadrane · sheet 4 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG

Process: print standing on the shoe, 0.16 mm; no supports (12 mm bridge over the 37 mm $\varnothing$ hole)
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

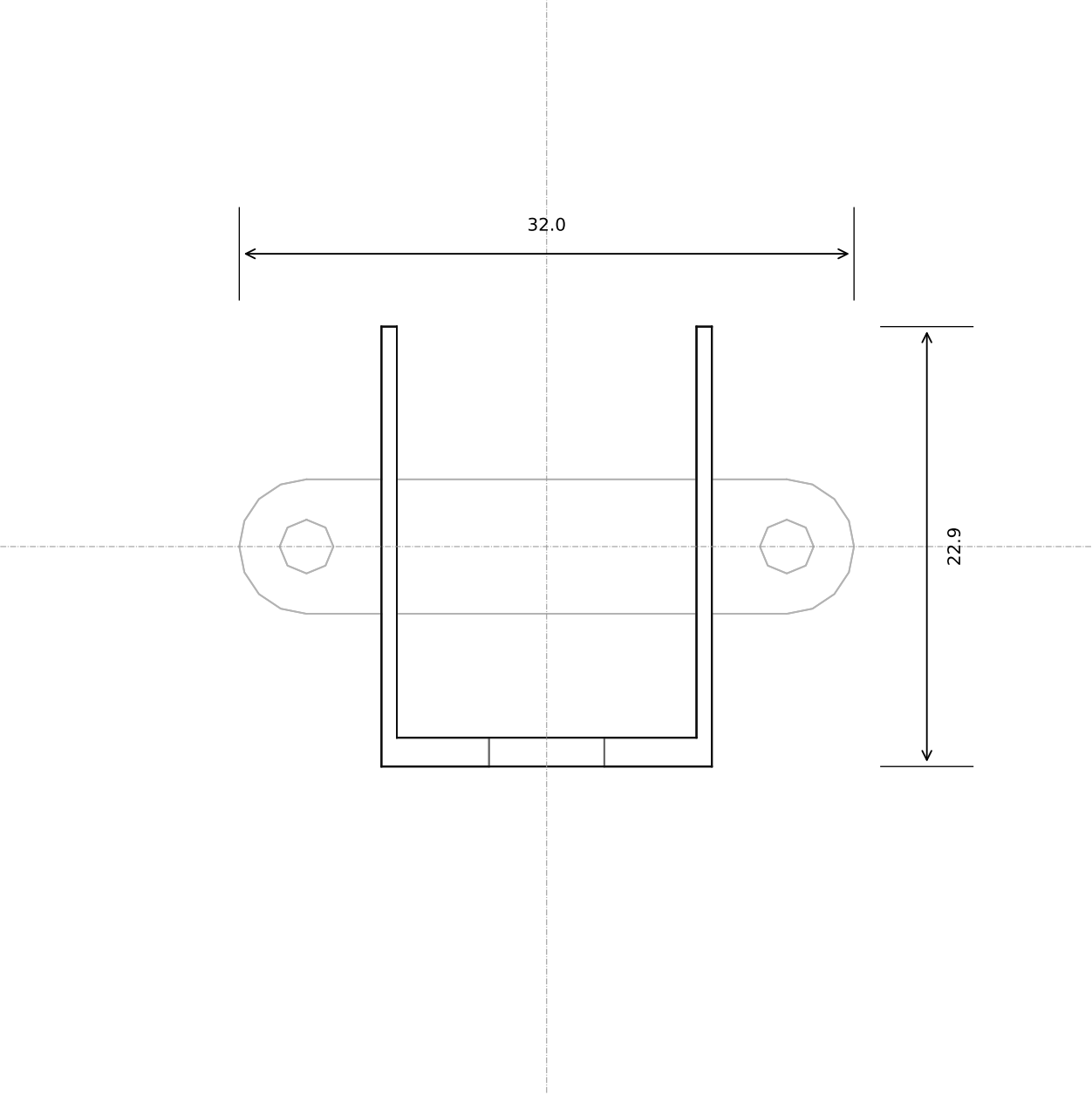
Part: carriage

Overall 37 mm $\varnothing$ x 40.8 mm, 14.5 perimeters for the tab

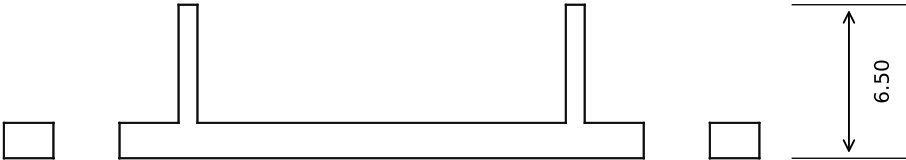
Volume 4.44 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)

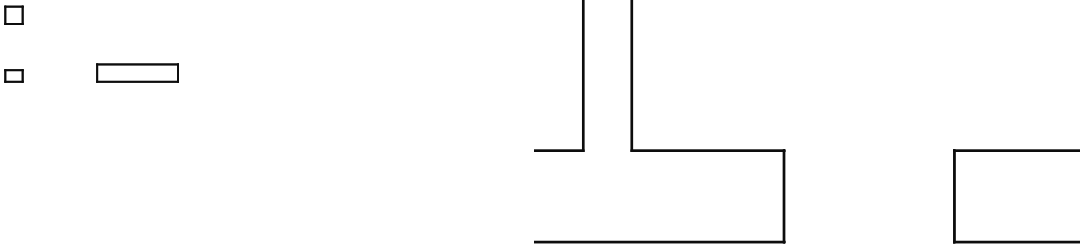


SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 8 to 18, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. floorless frame round the AGFRC envelope 21.4 x 15.2 x 6.0; walls 1.0, 6.5 tall; the servo lies on the plate at r 19.0-40.4, az 90
- 2. open toward the carriage tab; lead notch in the inner end wall
- 3. 2 ears at t ±12.5, M2.5 x 4 into tapped blind holes in the plate's web

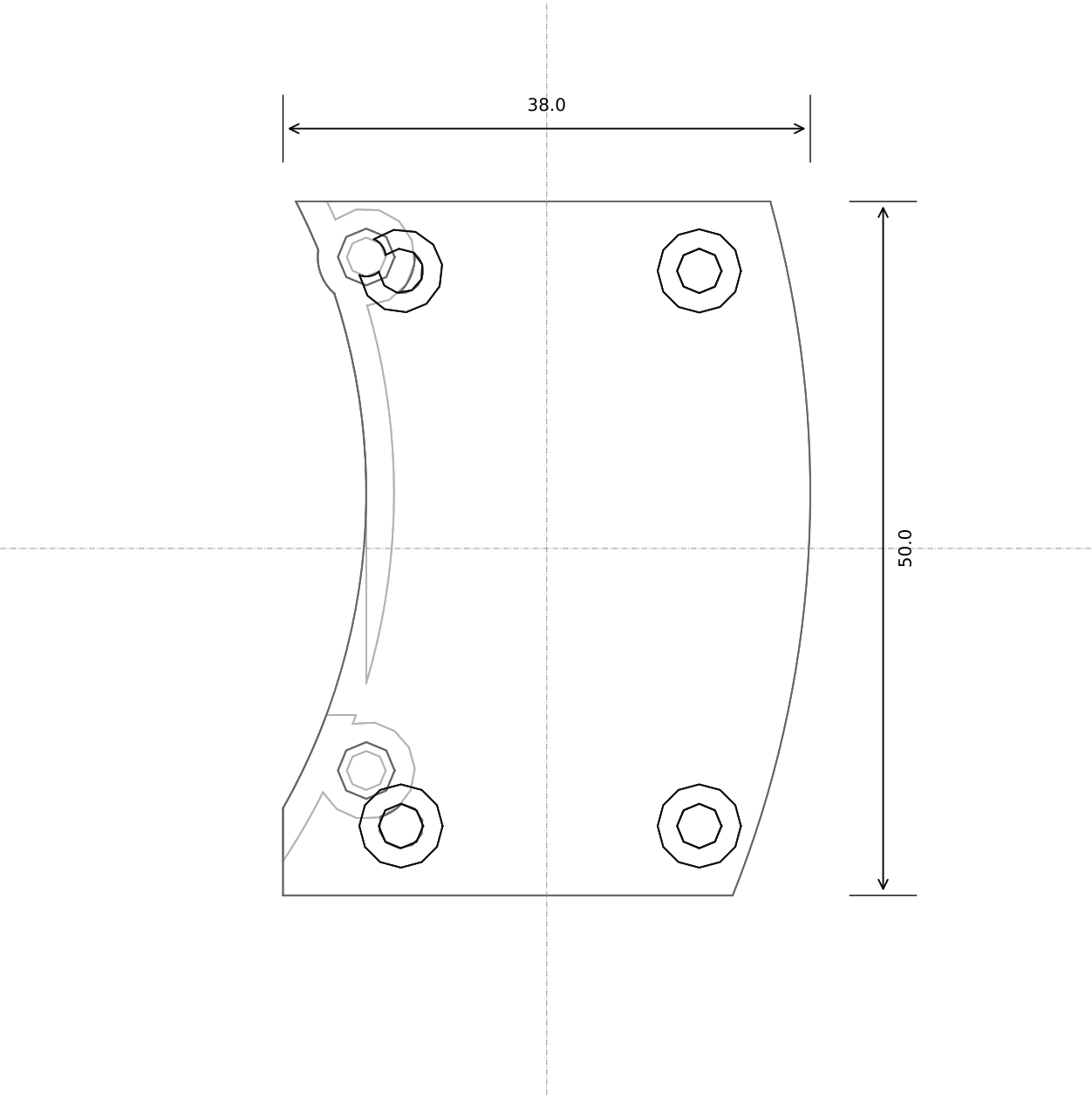
the 60 — v15 Servo frame (on the plate)

Cadrane · sheet 5 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

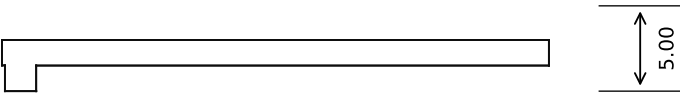
Material: PETG
Process: print flat, 0.20 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: servo_mount
Overall 32.0 x 22.9 x 6.5
Volume 0.64 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 10 to 21, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. slab 1.5 thick at z 9.5-11.0, r 46-78, t -29..21, bridging the port slot; its outer end rests on the port face's rail (z 9.5)
- 2. foot strip r 46-48 on the plate; two csk ears at [(46.0, 17.0), (46.0, -20.0)], M2.5 into the web
- 3. four Ø6 bosses 2.0 tall with M2 inserts for the connect board (30.0 x 46.0, holes 3 in from its inner corners, 5.5 from its outer)
- 4. the -y inner corner cut back to x 45.5 (the audio board's USB plug is there)

the 60 — v15 Connect-board bracket

Cadrane · sheet 6 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG

Process: print standing on its straight -y edge (slab vertical, feet and bosses clear by 38.0 x 20.0 x 5.0 supports)

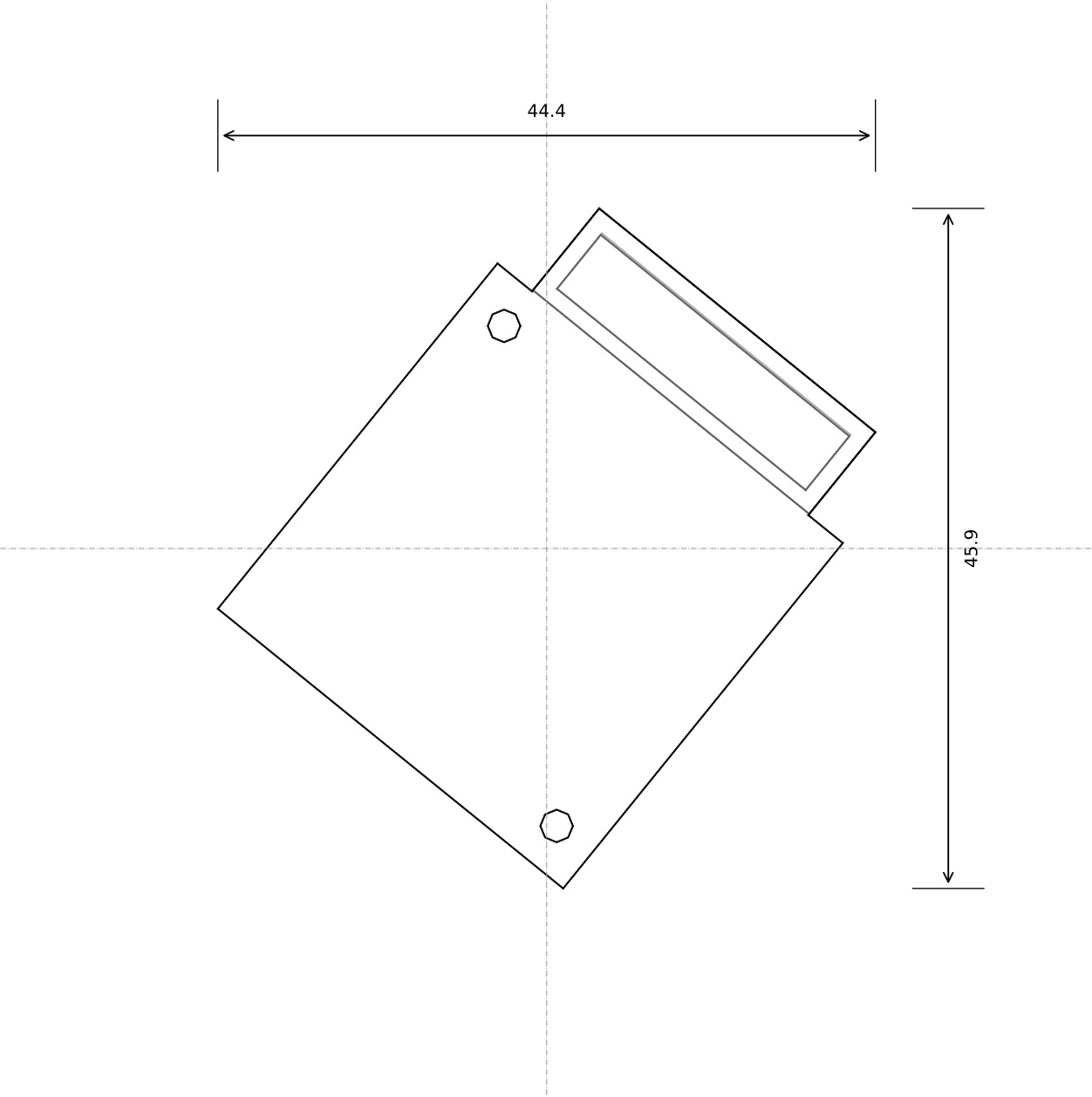
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: connect_bracket

Volume 2.76 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 24, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. plate 30 x 30 x 1.2 on the blower's top (z 18.3-19.5); four Ø5 x 0.5 pads for the motion board at ±12; two Ø2.2 through-holes on the blower's 24 x 24 diagonal: M2 x 18 pads through the board, the plate and the blower into the plate's piers
- 2. hood neck: 24 wide, r 66-73.2, walls 1.2, open at the bottom over the trench and toward the blower's outlet (21 x 7); on a 0.3 foam gasket on the plate top
- 3. blower at az 51, r 51 (Delta BFB0305HA-C 30 x 30 x 10, inlet down over the Ø22 hole in the web)

the 60 — v15

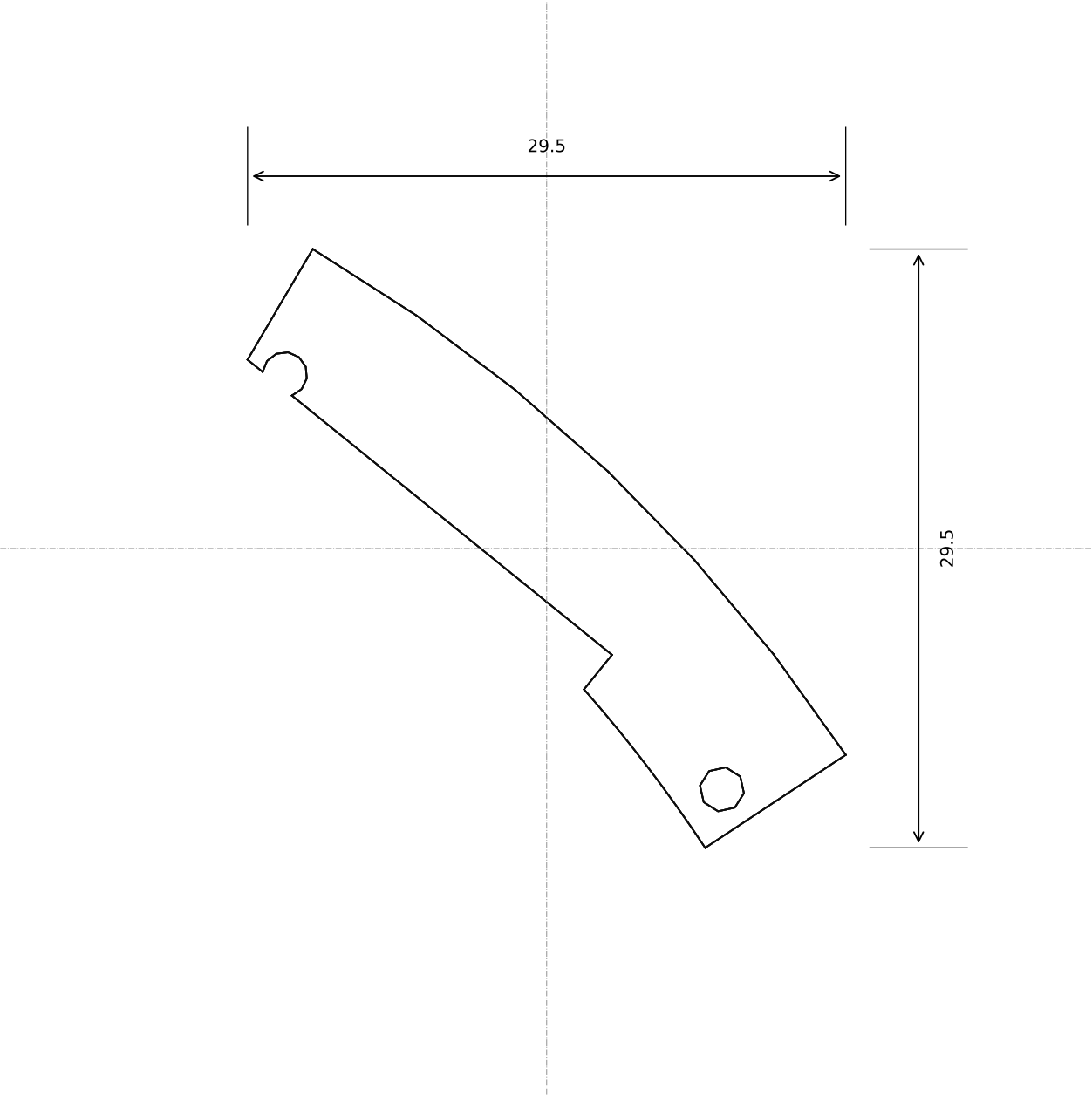
Blower saddle + hood neck

Cadrane · sheet 7 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v15: print on its top face (the motion board's face), 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: blower_saddle
Opens 44.4 x 45.9 x 11.2
Volume 2.10 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 7 to 17, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. 2.5 thick at z 8.3-10.8, az 33.5-59.5, r 72.2-80.5; butts the saddle's neck (0.15 clear); two Ø2.2 at az [35.0, 58.0], r 74.5: M2 x 5 into the ring's top; on the same 0.3 gasket

the 60 — v15

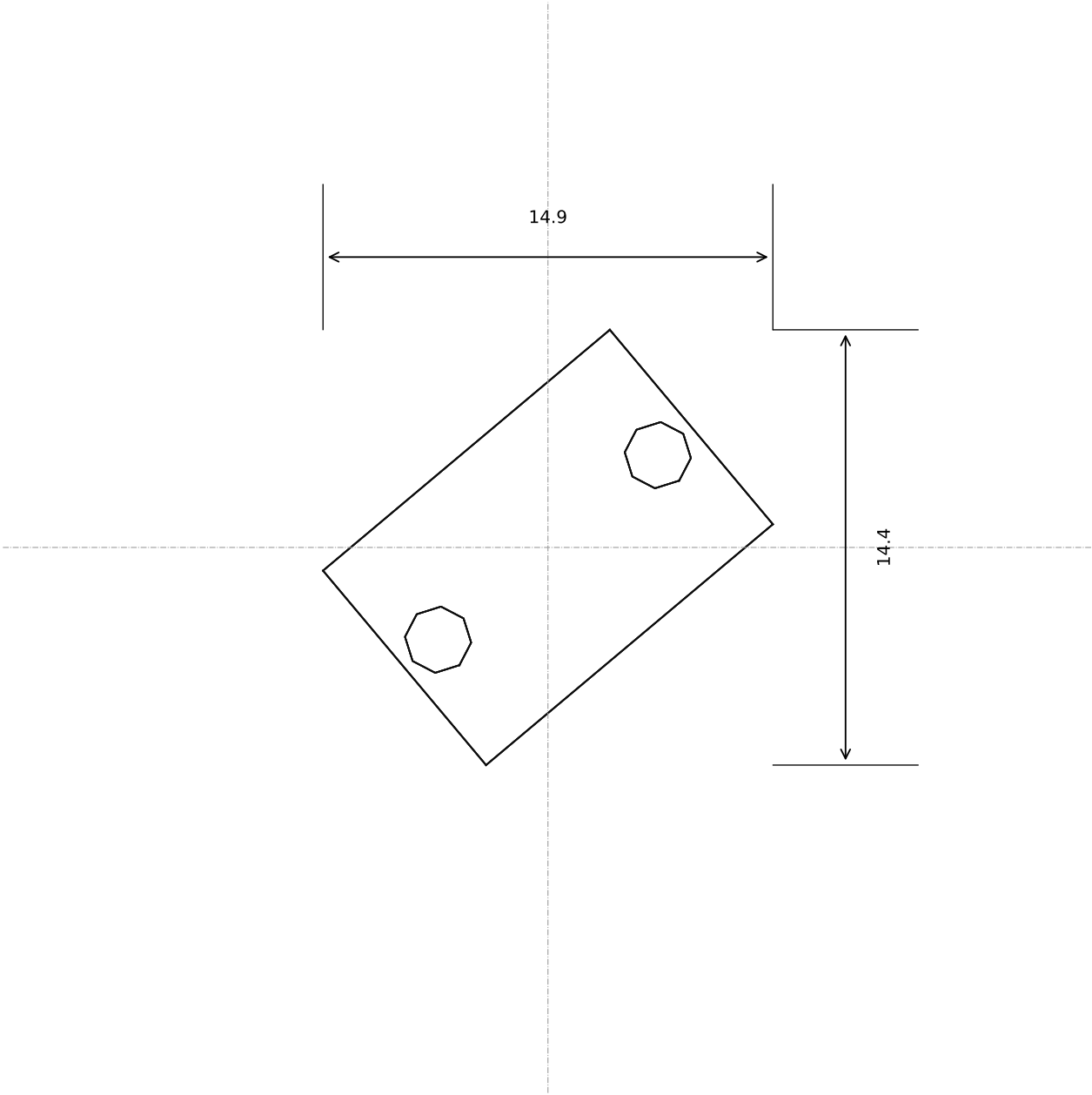
Hood lid (over the trench foot)

Cadrane · sheet 8 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

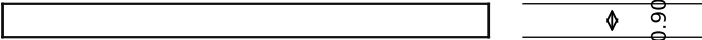
Material: PETG
Process: v15: print flat, 0.16 mm; no supports
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: hood_lid
Overall 29.5 x 29.5 x 2.5
Volume 0.61 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 4 to 9, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. 8.4 x 12.4 x 0.9 (nominal; family [0.7, 0.8, 0.9, 1.0, 1.1]), two Ø2.2 at ±4.75; sits on the seat flange under the encoder board, sets the 2.0 optical gap

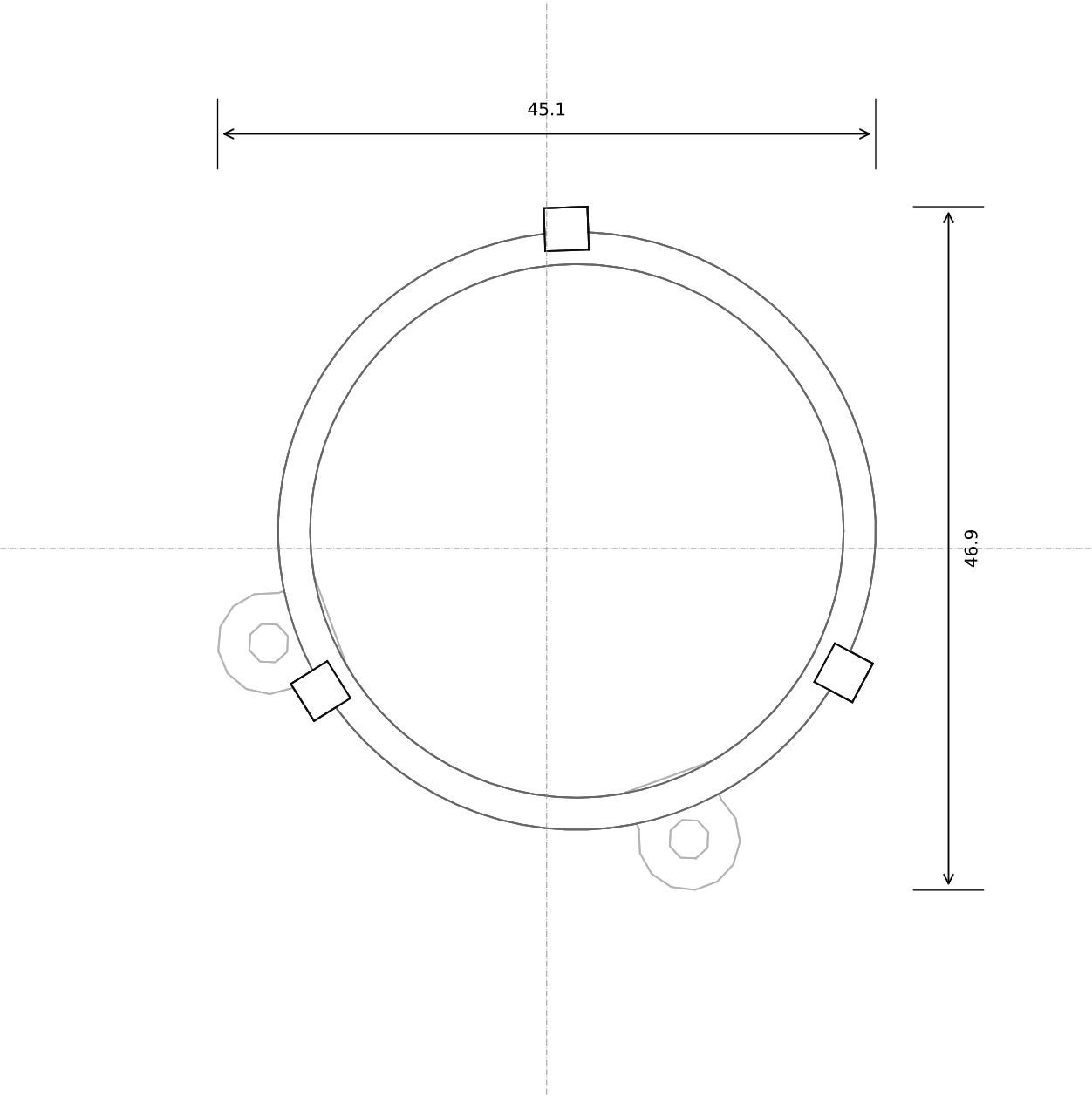
the 60 — v15Encoder shim (family)

Cadrane · sheet 9 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG
Process: v15: print flat, 0.10 mm; one of each height in ENC_SHIM_FAMILY
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: encoder_shim
Overall 14.9 x 14.4 x 0.9
Volume 0.09 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 11 to 25, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. ring r 18.3-20.5 x 4.45 under the Ø40 flange
- 2. three snap fingers at [92.0, 212.0, 332.0]° from the speaker's centre (device azimuths)
- 3. two ears at r 22.5, [200.0, 290.0]° from the centre; M2.5 x 4 from above into the plate's web (v11: M3 from below)
- 4. speaker centre (-36, 47) - az 127°, r 59.2

the 60 — v15

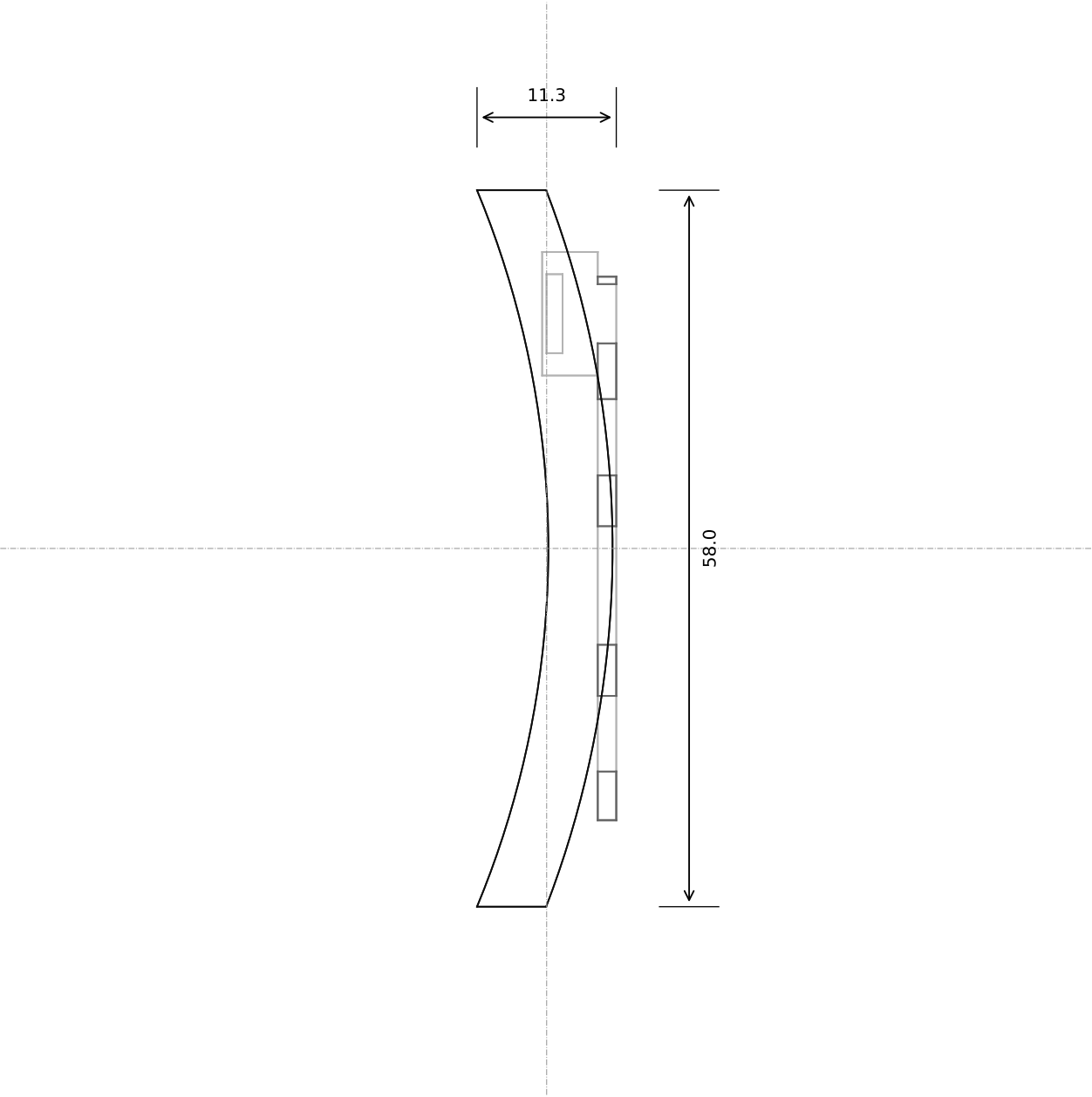
Speaker cradle

Cadrane · sheet 10 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

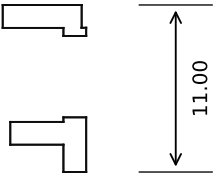
Material: PETG
Process: print flat, 0.20 mm; no supports (fingers full height)
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: speaker_cradle
Overall 45.1 x 46.9 x 7.2
Volume 1.44 cm³
Scale: see each view (fit)

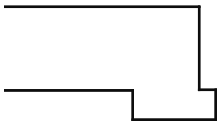
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



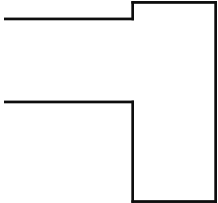
SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 8, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. 44.0 wide x 1.5 thick at r 79.704, z -1.5-8.0
- 2. barrel nose Ø6.2 at t -15.0, z 3.5; USB-C opening at t -3.0, z 4.5; jack Ø6.3 at t 9.0, z 4.6; light aperture Ø4.8 at t 19.0
- 3. shelf for the USB-C board; foot for the light-sensor board; rail along the top on the plate (1.5 tall, 58 wide, inner edge r 75.7, flat front in the face's plane, corners cut 45° to stay inside the diffuser) with two Ø2.2 holes at r 78.2, t ±26: M2 screws into the plate

the 60 — v15

Rear port face

Cadrane · sheet 11 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG

Process: print lying on its outer face, 0.16 mm; no supports (the socket holes are printed on the inner face)

mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

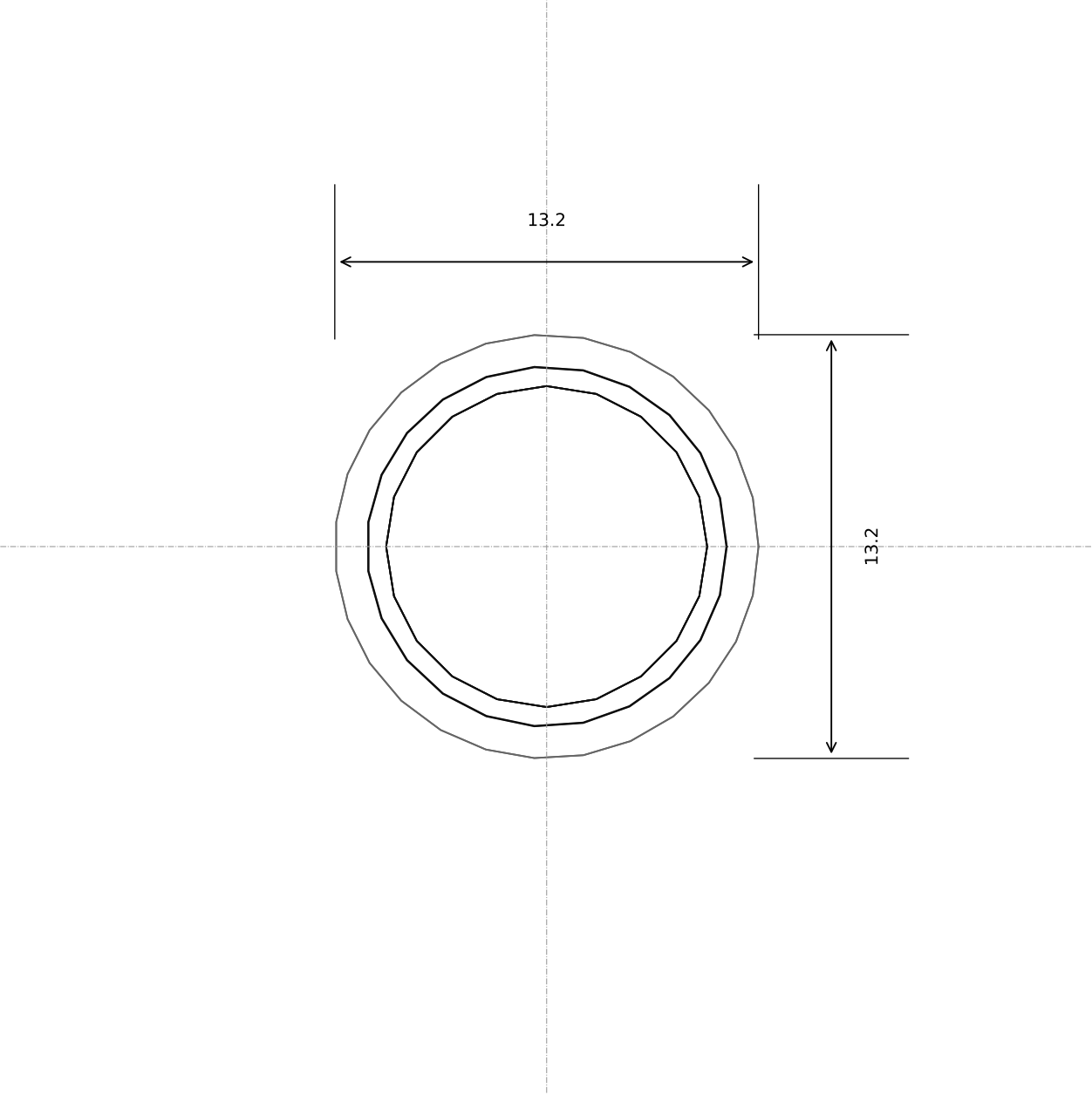
Part: port_face

Overall size: 58.0 x 11.0

Volume 1.01 cm³

Scale: see each view (fit)

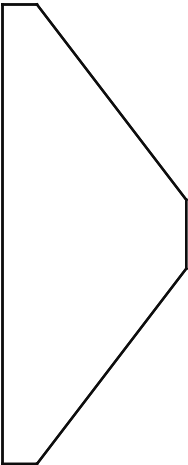
PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 3 to 9, enlarged)



SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø13.2 x 4.0, 90° V 1.3 deep with a 0.6 flat; bore Ø10 press over a 623ZZ

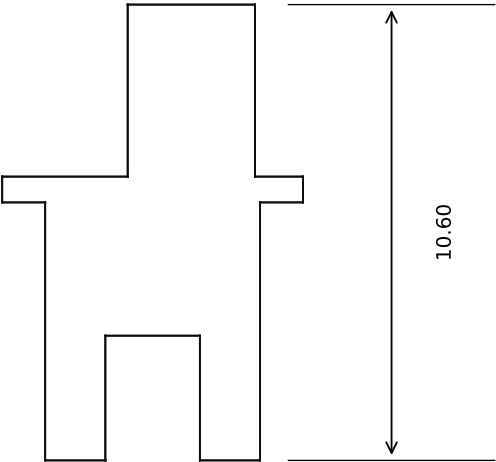
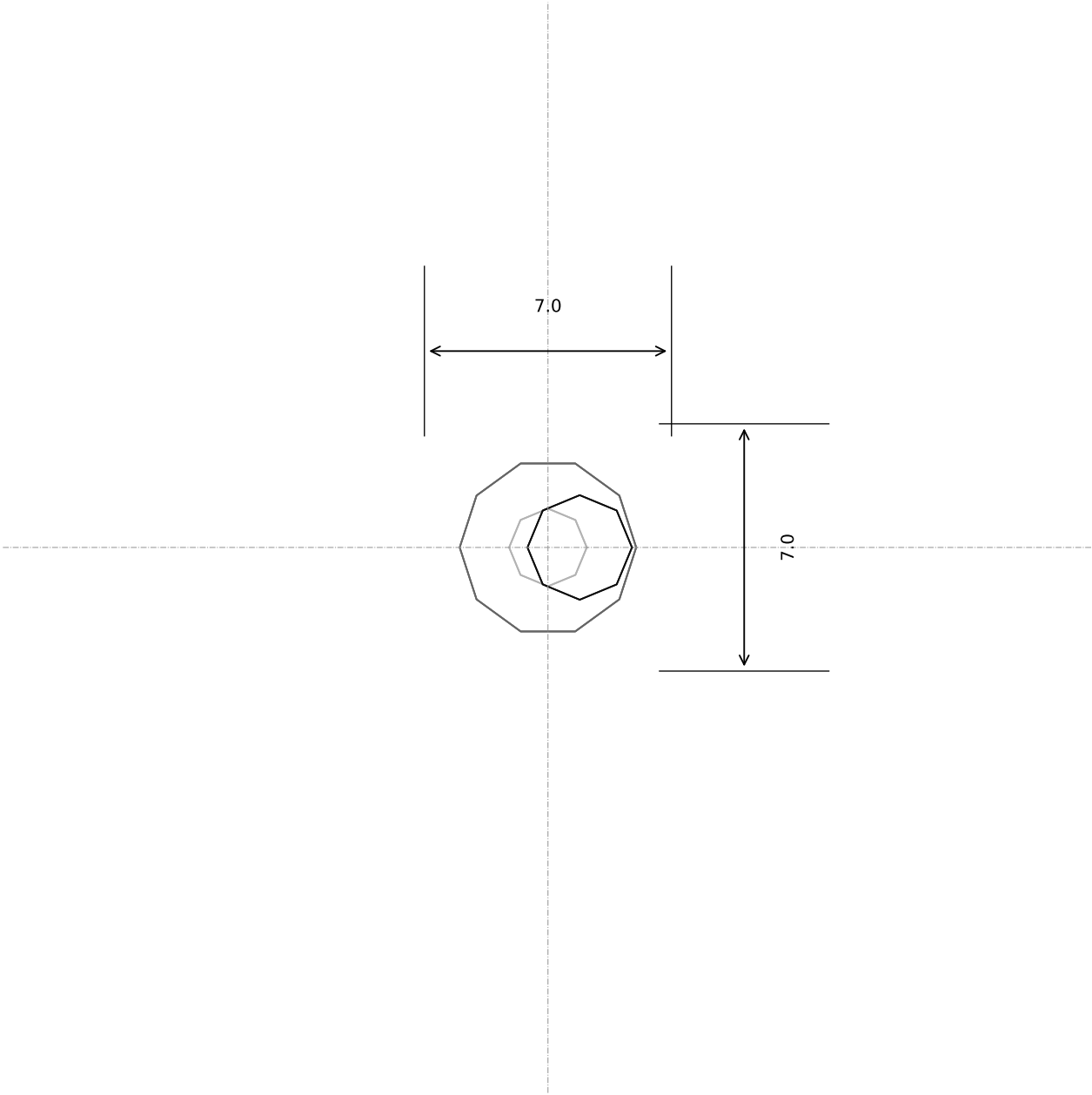
the 60 — v15 Wheel V-collar (x3)

Cadrane · sheet 12 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / POM (production)
Process: print axis vertical, 0.12 mm
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

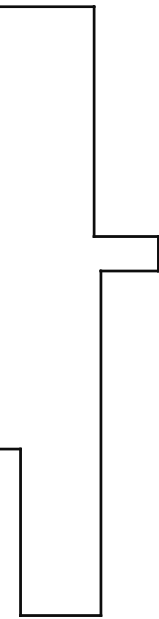
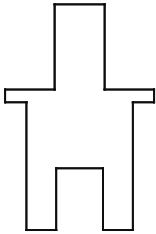
Part: collar_0
Overall 13.2 x 13.2 x 4.0
Volume 0.15 cm³
Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



DETAIL C (section A-A, x 2 to 6, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

1. Ø5.0 shank x 6, Ø7.0 x 0.6 flange, Ø3.0 pin x 4.0 offset 0.9; 2 mm hex socket

the 60 — v15

Eccentric bush (x3)

Cadrane · sheet 13 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: PETG (prototype) / brass (production)

Process: print stem down / pin up, 0.12 mm; the 1 mm flange ledge is the overall flange 7.0 x 10.6

mm. Tolerance ±0.1 unless stated; clearances: running 0.30,

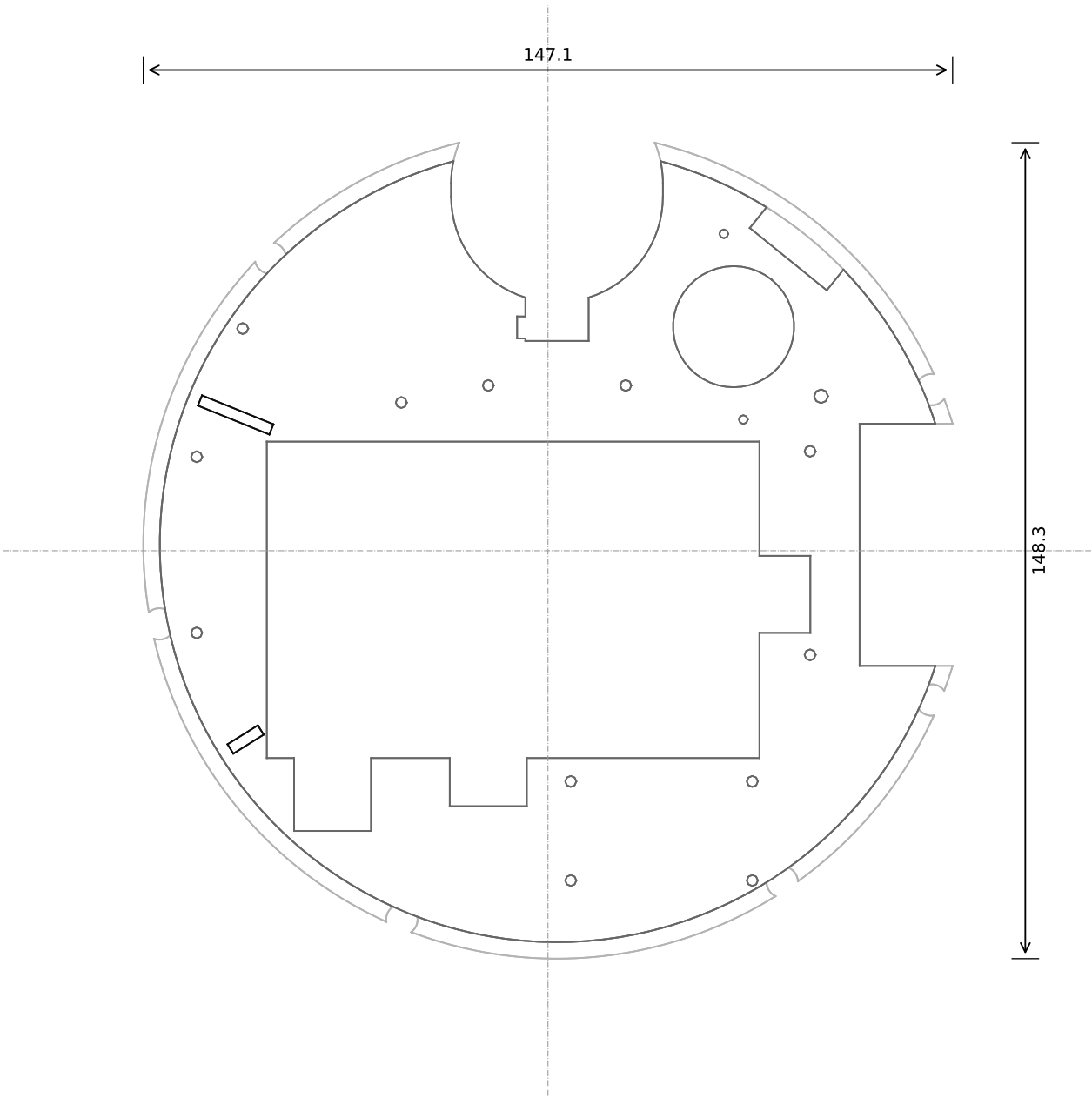
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: bush_0

Volume 0.16 cm³

Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 44 to 76, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

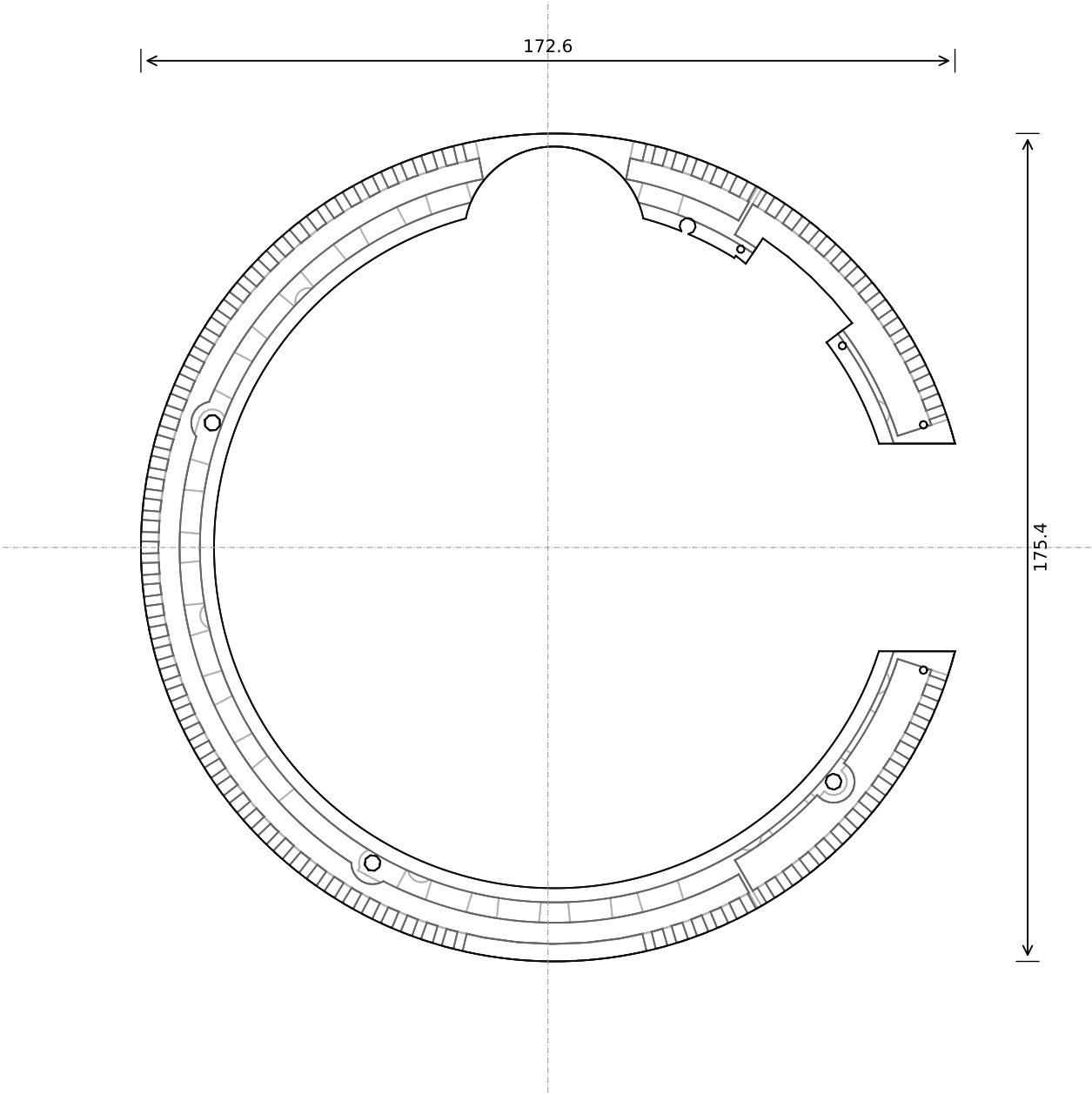
1. $r < 75.2 \times 8.0$: 2 web (z 6-8), 5 duct (z 1-6) closed by a 1 mm plate in a recess on a 0.2 gasket in a groove; $\varnothing 7$ piers 2.5 into the duct under every screw from the top; rebate r 72.2-75.2, z 5-8 for the ring's flange on a 5 shoulder, 7 x M3 csk from below at r 73.7; the ring's bond screw M3 tapped 3.0 into the shoulder at az 67.
2. duct: collector r 66-70.2 (front and sides), 19 straight channels 3 x 5 on a 4 pitch beside the Pi window, rear plenums r 55+; 2.5 walls round every through-cut; intake passages as closed tunnels z 1-4.5 through the duct wall and shoulder (17 x 9 wide) + 3 exhaust tunnels 10 wide at az [311.0, 320.0, 329.0]
3. BLOWER (v15, fitted): $\varnothing 22$ inlet hole through the web at az 51, r 51; discharge trench 18 wide from r 66.8 out through the duct wall and the shoulder, widening to az 37-56 (open-topped, the hood seals it); the 2 mm wall at r 64.8-66.8 keeps it off the suction side.
4. Pi window (Pi at (-52, -38), with the USB-C plug head, HDMI plug and USB-A plug notches), carriage hole, port slot; 12 x M2.5 tapped blind 3.5 deep from the top (converter audio, servo frame, connect bracket, speaker cradle) + 2 x M2 (the blower) + THE chassis bond (M3) at (48.0, 27.0), each on a pier; 9 x M2.5 tapped from the bottom.
5. 2 ribs 2.0 x 6.0 tall on the top face at r 56-70, az [158, 212]

the 60 — v15

Base plate - aluminium core (the heatsink)

Material: 6082 aluminium, CNC two-sided; EXTERNAL faces black hard anodised; INTERNAL faces chromate or bare - no masked pads (design for laser cutting)	Part: base plate
Process: machined, not printed (print a PETG stand-in without the duct for to verify fit)	Overall 147.1 x 148.3 x 14.0
mm. Tolerance ± 0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Volume 40.20 cm ³
	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



8.00

DETAIL C (section A-A, x 56 to 88, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

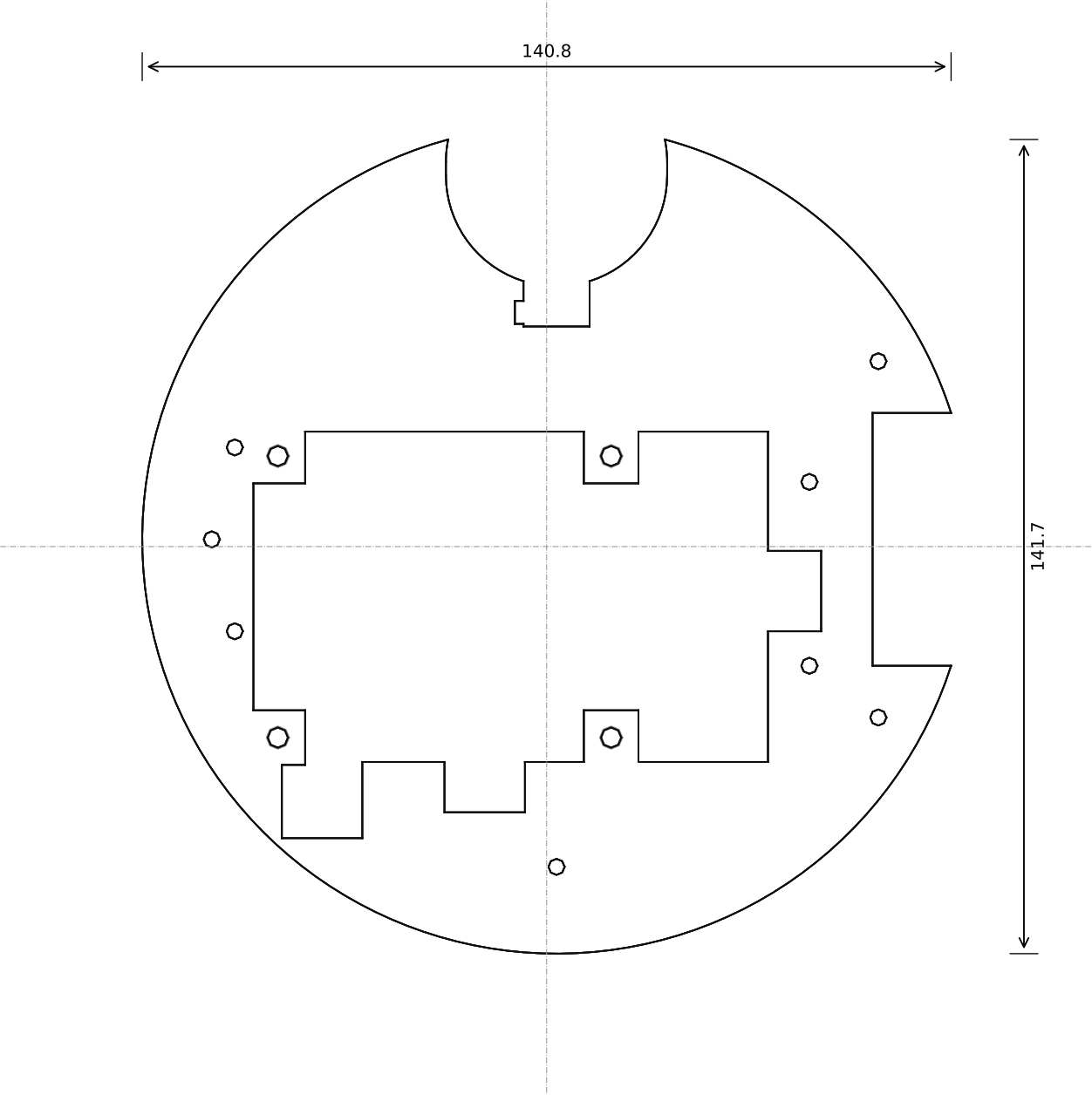
- r 75.2-87.7 x 8.0; 92 obround openings 2.0 x 4.5 (z 1.55-6.05) on 3° through the 3.7 outer wall, az 1.5 + 3k, mirrored about 0-180, none across the port face (342-18); 64 intake (60-300) + 28 exhaust (18-60, 300-342); 0.3 x 45° polished at every mouth; the upper land plain (no reeding - Ryan, 7 Sep); 0.3 chamfers on both sides of the openings; 0.3 chamfers on the outer wall (r 84.0-87.7) opens into it all round both arcs; the blower's trench: flange notched and inner land cut away az 37-56 to r 79.0, the groove's root at az 67.5, r 73.7 (M3 from the top into the core, star washer under the head, both faces bare)
- underside groove behind the openings: r 79.5-84.0 (intake arc) / r 76.5-84.0 (exhaust arcs), z 0-6.55, walls 1.2° at 60 and 300; the pad closes it; the 0.8 undercut under the outer wall (r 84.0-87.7) opens into it all round both arcs; the blower's trench: flange notched and inner land cut away az 37-56 to r 79.0, the groove's root at az 67.5, r 73.7 (M3 from the top into the core, star washer under the head, both faces bare)
- inward flange r 72.2-75.2, z 5-8, 7 x M3 tapped blind from below at r 73.7, az [22.5, 78.75, 135, 191.25, 247.5, 303.75, 337.5]; the dedicated BOND screw's Ø3.4 clearance at az 67.5, r 73.7 (M3 from the top into the core, star washer under the head, both faces bare)
- passages across the inner land (r 75.2 to the groove), 3.5 deep, the pad closes them: 17 intake 9 wide (536 mm²) on the 11.25° grid, 3 exhaust 10 wide at az [311.0, 320.0, 329.0] (the -y circuit; the +y circuit exhausts through the trench)
- 3 x M3 csk from below at r 77.2, az [160, 240, 320] (the structure's pillars); 2 x Ø1.6 tapped M2 at r 78.2, t ±26 (the port face's rail); port slot 44.0 wide

the 60 — v15

Rim ring (stainless)

Material: stainless steel 304/316, machined; BARE: electropolish (the 0.3 chamfers running clockwise, then brush the chamfers running clockwise)	Overall 172.6 x 175.4 x 8.0
Process: machined, not printed	Volume 26.17 cm ³
mm. Tolerance ±0.1 unless stated; clearances: running 0.30, static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.	Scale: see each view (fit)

PLAN (from above): sections at three heights — bottom (light), middle, top (dark)



SECTION A-A (front, plane y = 0; x across, z up)



DETAIL C (section A-A, x 40 to 72, enlarged)

SECTION B-B (side, plane x = 0; y across, z up)



Key dimensions (from the parameter block):

- 1. disc r 72.1 x 1.0, flush in the core's recess on a 0.2 die-cut gasket; same Pi window, carriage hole and port slot as the core
- 2. four 9.5 square lands reaching into the Pi window at the Pi's holes [(-48.5, -34.5), (9.5, -34.5), (-48.5, 14.5), (9.5, 14.5)], Ø2.7 csk from below: the Pi and adapter stack screws to them (M2.5 x 6 into the standoffs)
- 3. 9 x M2.5 countersunk; unscrew to clean the fin channels

the 60 — v15

Closing plate (aluminium)

Cadrane · sheet 16 of 16 · 2026-09-06 · drawn from src/params.py + model.py (build123d)

Material: 1 mm aluminium, laser-cut, black anodised (outer face)
Process: laser-cut, not printed
mm. Tolerance ±0.1 unless stated; clearances: running 0.30,
static 0.15, knob 0.40. Printed as-built; CNC knob Ra 0.8 on the chamfers.

Part: closing_plate_AL
Overall 140.8 x 141.7 x 1.0
Volume 9.31 cm³
Scale: see each view (fit)